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APPLIED RESEARCH

Optimizing Energy and Spectral Efficiency in Mobile Networks: A Comprehensive Energy Sustainability Framework for Network Operators

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ABSTRACT The urgent need to meet Environmental, Social, and Governance net-zero commitments and the financial risks posed by rising energy costs, are placing increasing pressure on Mobile Network Operators (MNOs) to optimise energy use. Given the significant energy consumption from Base Stations (BSs) and the fact that Fifth Generation (5G) deployments are not meeting expectations for reduced energy consumption when compared to Fourth Generation (4G), MNOs are shifting from a load-centric optimisation approach to a service-centric paradigm that balances spectral efficiency and energy efficiency. This paper introduces the Energy Sustainability Framework, which provides a composite index to evaluate the spectral and energy efficiencies of BSs, summarised across four classes: A, B, C, and D. This framework supports network auditing, benchmarking, monitoring, and energy efficiency optimization while ensuring the desired Quality of Service (QoS). By leveraging Machine Learning (ML) and Explainable Artificial Intelligence (XAI) techniques applied to live data - including Performance Management (PM) and Energy Management (EM) metrics - integrated with domain expertise, the framework goes beyond deterministic computation to identify the root-cause factors influencing the BSs' energy class, thereby enabling targeted optimisation. The framework can be flexibly applied at the location (site), BS (cell), or equipment type (hardware) levels. Notably, the implemented ML and XAI model achieved an F1-Score of 0.78 for 5G scenarios and 0.83 for 4G scenarios in predicting the Energy Sustainability Class, relying on a reduced set of indicators encompassing the most influential factors. An additional analysis, based on domain expertise, provided evidence of potential causal relationships between the class and observed variables involving channel quality, Multiple Input Multiple Output (MIMO) utilisation, network topology and energy-saving functionalities. For example, the results show that using higher order modulation coding schemes reduces the probability of a BS falling into class D by 50%. These findings provide practical insights for developing more sustainable network operations.

INDEX TERMS Explainable artificial intelligence, machine learning, wireless networks, 5G, energy efficiency, spectral efficiency, sustainability.

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I. INTRODUCTION

In recent years, the Energy Efficiency (EE) of Base Stations (BSs) emerged as a primary concern for the Mobile Network Operators (MNOs). This resulted from the escalating energy

costs [1] and the global imperative to combat climate change, primarily driven by CO₂ emissions [2]. The global emissions of CO₂, caused by fossil fuels and industry, have reached a historical high of 37.6 gigatons in 2023, surpassing the pre-pandemic emission levels [2], [3].

Failure to reverse this upward trend within the next decade, poses a severe threat to natural ecosystems on our planet [4]. For instance, Europe has set an objective to achieve net-zero emissions by 2050 [5]. Moreover, the increasing energy cost is a significant business risk for the MNOs, potentially impacting their overall financial performance [6]. Despite not being a major contributor compared to other industries, the telecommunications sector accounted for approximately 3% to 4% of global emissions in 2022 [6], [7]. These emissions can be split into three scope groups, with the respective internal contributions: 1) direct emissions (4%); 2) indirect emissions caused by purchased energy (25%); 3) indirect emissions caused by other purchased services and products (71%). Over the past decade, tier one MNOs such as Vodafone, Orange, BT and T-Mobile achieved reductions of over 50% in their indirect emissions caused by purchased energy [4]. These efforts resulted from the shift towards renewable energy sources. For example, the Vodafone group reported that between 2022 and the first half of 2024, their worldwide percentage of purchased electricity from renewable sources changed from 69% to 84%; and in Europe, this percentage reached 100% in 2023 [8].

In mobile networks, about 75% to 80% of the energy consumption originates from BSs [4], [9]. Both environmental and financial considerations would advise against proliferating BSs to contain energy consumption. However, industry demand exerts opposing pressures due to the ever-increasing customer traffic, which is expected to grow until 2029 by over 20% annually [10]. Moreover, despite the promises of deploying Fifth Generation (5G) while reducing total network energy consumption [11], initial field results indicate the opposite. A survey conducted on 65 MNOs revealed that 69% of them reported 5G as being less energy efficient than Fourth Generation (4G) in the short term [12]. These concerns are echoed in [13], highlighting the lack of publicly available evidence regarding the actual energy use implications of 5G.

Furthermore, the traditional load-centric optimisation approach of MNOs led to deploying new technologies on top of existing ones, resulting in excessive energy use. This underscores the increasing need to prioritise energy optimisation throughout the network life cycle [14], [15]. One immediate consequence is the increased adoption of Radio Access Network (RAN) sharing initiatives and neutral host operators, facilitating multi-tenancy at BS locations [15]. Additionally, Second Generation (2G) and Third Generation (3G) mobile networks are gradually being switched off, while 4G and 5G radio equipment is optimised through shared configurations [16], [17]. These actions aim to rationalise network architectures and reduce the

burden of legacy systems and redundant infrastructures on energy consumption. Looking ahead, transitioning to a service-centric model requires the combined optimisation of allocated resources - such as energy consumption - while meeting specific service demands. For example, studies such as [18] and [19] highlight the challenges MNOs face in adopting a service-oriented paradigm, including issues related to network security, cost efficiency, Quality of Service (QoS), and energy consumption.

This paper introduces the Energy Sustainability Class (ESC), a composite index that assesses both Spectral Efficiency (SE) and EE of BSs, grouping their performance into four classes (A, B, C, and D). Furthermore, the ESC concept is expanded into a comprehensive Energy Sustainability Framework (ESF), combining Machine Learning (ML) and Explainable Artificial Intelligence (XAI) techniques with domain expertise to optimise the BSs' class by identifying its root-cause factors. The proposed framework offers several advantages. Firstly, the ESF introduces the ESC, which denotes the class of each BS, serving as a unified metric for optimising both SE and EE, balancing QoS with energy consumption. Secondly, due to its normalised structure, the ESC supports MNOs in regional benchmarking, helping assess optimisation efforts, equipment swaps, and RAN provider comparisons. Thirdly, the ESF is designed to be independent of the chosen ML model, Radio Access Technology (RAT), and RAN equipment provider, allowing application at different aggregation levels, such as the BS, location, or equipment type. Lastly, it provides interpretable results to support optimisation decisions, identifying the input features contributing most to the ESC and quantifying their relative importance.

The main contributions of the paper are then the following:

- Proposal of the ESC, a composite index that categorises a BS into four classes - A, B, C, and D - based on SE and EE.
- Development of the ESF, a framework using a ML model enriched with XAI capabilities to assess the most influential indicators impacting the ESC of BSs.
- Discussion on the ESC optimisation root-cause factors, relying on a mix of domain expertise and ML/XAI results based on data from a live 4G/5G network.

The paper is organised as follows. After the introduction in Section I, related work is reviewed in Section II. Section III introduces the ESC concept and describes its theoretical background, while Section IV outlines the complete ESF to optimise the ESC, comprising five steps. The data considered in this paper is detailed in Section V. In Section VI, the ESC is computed for each BS, and in Section VII a ML model is introduced to evaluate the factors conditioning the observed classes. Section VIII explains the feature selection process to identify key features impacting the ESC, while Section IX examines root-cause factors that can be used to optimise this metric. Finally, Section X presents the main conclusions and outlines future work.

II. RELATED WORK

This section provides an overview of related work, highlighting the technology enablers for optimising the energy efficiency of mobile networks. Special attention is given to the dual challenge of controlling operational costs while maintaining the required QoS for end users. Previous works aimed at enhancing the EE of multi-RATs mobile networks are then discussed, highlighting the role of Artificial Intelligence (AI)/ML in driving sustainable energy consumption across legacy RATs like 4G, alongside 5G, and the upcoming Sixth Generation of cellular networks (6G) [20].

A. TECHNOLOGY ENABLERS

Given the critical need to reduce energy consumption costs and carbon emissions, various approaches, including technologies and procedures, have emerged as key enablers to support the MNOs in transforming their networks towards greener 5G and beyond mobile networks. In [9], the authors identified six areas of improvement on the road to 6G that will result in energy consumption reductions: 1) power amplifier improvements; 2) SE increase; 3) reduced signalling; 4) advanced sleep modes; 5) network virtualisation; and 6) AI native incorporation. Along the same lines, in [4], four key development areas are outlined for RAN EE optimisation: 1) massive Multiple Input Multiple Output (MIMO); 2) lean carrier design; 3) sleep modes and 4) ML. Notably, as a technology-agnostic framework, the ESF proposed in this paper can serve as an optimisation tool for any pervasive technology, provided sufficient data is available to compute SE and EE.

B. ENERGY EFFICIENCY OPTIMISATION

The importance of ensuring a fair trade-off between controlling energy consumption costs and maintaining QoS was recognised in early works, such as [21], [22], and [23]. These studies analysed the impact of wireless channel characteristics, including fading, path loss, and interference, in conjunction with various deployment strategies, notably involving small cell deployments and massive MIMO configurations, to achieve a balance between SE and EE in cellular networks.

The problem of optimising energy consumption in heterogeneous networks considering factors such as QoS, SE, EE, and latency was also analysed in [24], where an algorithm is proposed to optimise the cellular association to the macro or small-cells layer, based on the EE. In [25], the problem of maximising the SE and EE was studied through the joint optimisation of power control and access point association and selection. Along the same lines, in [26], the maximisation of the overall SE and EE, subject to user-specific QoS requirements, was analysed by jointly optimising user pairing and power allocation. A more comprehensive approach is given by [27], where the problem of optimising 5G and 6G networks integrates Key Performance Indicators (KPIs) from different layers of the network, aiming at reaching

solutions that account for the energy consumption of the BSs, the total capacity provided to the end users, and also the signalling cost generated by handovers. In [14], the authors point out that the continuous deployment of new technologies alongside existing ones, following a load-centric philosophy, has led MNOs to increased energy use and emissions. Alternatively, a service-based architecture is recommended for multi-RAT cellular networks. In [4], two optimal network operating points are derived for both the downlink and uplink directions, considering the trade-off between SE and EE: 1) the optimal transmit power for BSs and User Equipments (UEs); 2) the optimal BS and UE density.

C. AI/ML-DRIVEN OPTIMISATIONS

Using AI and ML techniques for optimising energy consumption is a prolific research area. Moreover, given the surge in energy consumption costs after 2022 [1], the implementation of sleep modes has recently gained attention in both 4G and 5G networks. In essence, the sleep mode concept involves switching off radio units during periods of low utilisation. Specifically, integrating AI and ML techniques with radio access sleep mode functionalities enables dynamic activation and deactivation, relying on the network status, in contrast with hard-coded switch-on and switch-off triggers.

In [28], a comprehensive taxonomy of AI/ML applications in 5G networks for improving EE is provided. The taxonomy covers the hierarchical network layers, including access, edge, and core networks, while introducing a fourth dimension focused on energy harvesting. The access network connects end-user devices to edge networks using various RATs. The edge network brings computing resources closer to end-users, reducing latency and offloading traffic from the core network, thereby improving responsiveness and optimising bandwidth usage. The core network is the backbone, managing data routing, switching, and network control functions across the broader network. Energy harvesting involves capturing ambient energy from external sources - such as thermal, solar, wind, kinetic, radiation, and magnetic - which can be stored or directly used to power wireless devices.

An example of a ML application to manage the sleep mode activation is provided in [29], where a framework is proposed to dynamically adapt the sleep mode configuration to the actual traffic while considering the QoS constraints, notably the BS reactivation delay. In [30], a model is proposed to optimise the energy consumption of BSs by dynamically shutting down carriers. Similarly to the ESF, this approach leverages live network data and employs a blend of ML and specific domain expertise. A theoretical approach to optimise the activation of deep sleep mode is given by [31], where a Reinforcement Learning (RL) mechanism is proposed to find the best trade-off between EE and QoS constraints, notably introducing the latency requirements of ultra-dense 5G networks. In [32], the application of ML algorithms is explored across various facets in the context of Open Radio

Access Network (ORAN) architectures. This study highlights the crucial role supervised learning plays in intelligent network management and optimisation problems. In [33], a new Radio Resource Management (RRM) technique is proposed that integrates a per user Quality of Experience (QoE) criterion to manage the switching of small cells to sleep mode, when they are not serving active users, while ensuring the QoE requirements. In [34], live network data was employed to quantify the impact of various Performance Management (PM) indicators, including sleep mode activation, in the EE of BSs.

D. ADDITIONAL REMARKS

While some prior approaches have employed live network data, many rely on theoretical or simulation-based methodologies. Although they provide valuable insights, these approaches frequently overlook the complexities and variability inherent in real-world network environments. Notably, they often assume prior knowledge of user and UE distributions, which is seldom available in practice. As highlighted in [28], research on energy consumption optimisation using data from live wireless networks remains scarce. This disparity can be attributed to several challenges, including the inherent complexities of accessing live network data, managing its volume and variety, ensuring data quality and integrity, and addressing concerns related to security and privacy [35]. Additionally, the lack of real-time data extraction and labelling tools complicates the availability of relevant metrics for AI-driven network automation approaches [36]. This gap underscores the necessity for frameworks validated with live network indicators, such as the ESF.

III. ENERGY SUSTAINABILITY CLASSIFICATION

This section introduces the ESC, a novel concept that classifies a BS into one of four possible categories based on its energy sustainability performance. The main definitions associated with the ESC are presented, along with the required theoretical background and the respective calculation methodology.

A. CONCEPT OVERVIEW

The main idea behind the ESC concept is to generate a unique composite metric considering both SE and EE, offering an integrated perspective on these two often concurrent domains. This is achieved through a scatter plot representation of the SE and EE of BSs, where each BS is categorised into one of four classes - A, B, C, or D - based on the segmentation of SE and EE values. Fig. 1, illustrates the energy sustainability scatter plot, where the horizontal and vertical axes denote the SE and the EE, respectively. Each circle represents a BS, with its colour indicating its classification into one of the four possible classes defined by the ESC. The underlying definitions, theoretical background, and methodology are detailed in the following subsections.

Each considered class can be described as follows:

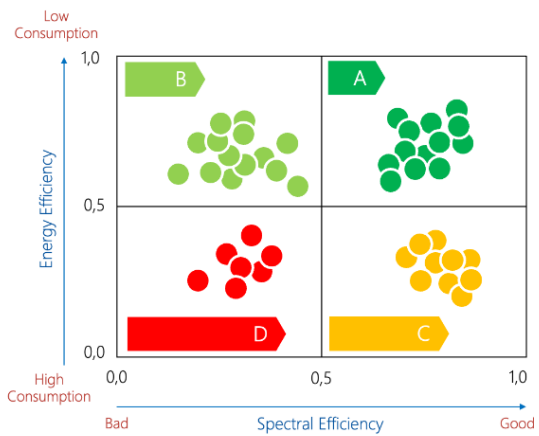


FIGURE 1. Scatter plot of ESC for different BSs, represented by circles with distinct colours in specific classes (A, B, C, D).

- **A** - The sustainable and desirable operating zone, with a good balance between SE and EE.
- **B** - The EE is good, but the SE is poor, denoting either very low utilisation, capacity problems or low QoS.
- **C** - Despite good SE, in this zone the EE is low, denoting a too high energy consumption.
- **D** - The unsustainable operating zone to be avoided, with low SE and low EE.

The ESC is independent of the technical characteristics of the RAT, equipment providers, or other factors, thus enabling comparisons between BSs, either within a given MNO's network or between MNOs, streamlining benchmarking analyses. Moreover, the ESC can be calculated for individual BSs or groups of BSs. This facilitates the identification of problematic BSs, and the development of optimisation actions to improve their performance and the respective energy efficiency.

B. DEFINITIONS

Some concepts are worth defining beforehand due to their frequent use throughout this paper:

- **Cell** - In a mobile network, a "cell" represents a geographic area served by a single radio transceiver, associated with a specific frequency band. Cells are the foundational units of the network, designed to efficiently manage radio frequencies, enhance network capacity, and enable seamless mobile connectivity.
- **Base Station (BS)** - In the context of this paper, a BS, denotes a group of cells deployed at a given network location, operating on particular RATs (4G, 5G or 4G/5G) and using a specific duplex mode (Time Division Duplex (TDD) or Frequency Division Duplex (FDD)). Moreover, active and passive equipment is required for the BSs' operation, notably the Baseband Units (BBUs), the Remote Radio Units (RRUs), and the antennas.
- **Spectral Efficiency** - It is measured by the ratio of data throughput to the available bandwidth. This indicator

represents the efficiency of the wireless channel in delivering satisfactory QoS to end-users, taking into account the available radio resources at each BS.

- **Energy Efficiency** - The ratio of the data volume to the energy consumption. This metric represents the efficiency of the wireless channel in providing the target QoS to end-users considering the used energy.

C. THEORETICAL BACKGROUND

Historically, MNOs have focused on optimising their BSs to achieve the highest possible data throughput, defined as:

$$\text{Avg_Throughput} = \frac{\text{Total_Traffic}}{\text{Transmission_Time}} \text{ [bit/s]} \quad (1)$$

where Total_Traffic denotes the transmitted bits in the downlink or uplink directions, while the Transmission_Time refers to the time spent on transmission. In this paper, calculations were performed per BS, with a minimum granularity of one hour. Moreover, MNOs typically assess their network's optimisation level by measuring the SE, given by:

$$\text{SE} = \frac{\text{Avg_Throughput}}{\text{Avail_Bandwidth}} \text{ [bit/s/Hz]} \quad (2)$$

where the term Avail_Bandwidth refers to the available bandwidth for users in a BS, considering either the downlink or uplink direction. Importantly, the maximum value of the SE at a given BS is determined solely by its RAT, which sets the upper limit on achievable throughput, depending on the available bandwidth. While this strategy effectively addresses the core aim of maintaining satisfactory QoS for end users, it overlooks the energy consumption required to deliver such a level of service. Given the escalating energy costs and the imperative to promote sustainable resource utilisation, MNOs are compelled to evolve from the traditional approach towards a balanced approach, which seeks the joint optimisation of the network capacity and energy consumption. This involves managing the right balance between SE and EE, defined as:

$$\text{EE} = \frac{\text{Avg_Throughput}}{\text{Power_Delivered}} \text{ [bit/s/W]} \quad (3)$$

where Power_Delivered reflects the total amount of power delivered to the RRUs serving the BSs. The EE can also be written as:

$$\text{EE} = \frac{\text{Total_Traffic}}{\text{Energy_Consumption}} \text{ [bit/J]} \quad (4)$$

where Energy_Consumption represents the total energy consumed by the RRUs in each BS, measured with a minimum granularity of one hour. The maximum achievable EE occurs in the extreme case where the maximum possible traffic is transmitted with the minimum energy consumption. The maximum traffic for each BS is determined by the maximum SE and the available bandwidth. Conversely, the minimum energy consumption is reached when the transmitted traffic per BS tends to zero.

D. METHODOLOGY

The calculation of the ESC for BSs requires a normalisation of the original SE and EE values, using their respective maximum achievable values, to ensure a fair assessment of energy sustainability performance across BSs and RATs. The maximum achievable SE for each RAT (*e.g.*, 4G or 5G) was calculated based on the RAT's technical specifications, as outlined in [37]. For each RAT, the calculation considers the highest possible values for parameters such as the Modulation Coding Scheme (MCS), the use of MIMO, and other relevant factors. Regarding the EE, the maximum achievable efficiency corresponds to the minimum energy consumption at the BS level, which can not be computed deterministically since it depends on specific vendor implementations. In this paper, an estimation for the minimum energy consumption in a BS for 4G and 5G was derived from [38], where a linear model using live network data was employed to predict the energy consumption of BSs.

Considering the RAT of each BS, the available bandwidth and the estimated value for the minimum energy consumption, the maximum values for the SE and EE, denoted by $\text{SE}_{\max}$ and $\text{EE}_{\max}$, respectively, are computed for each BS. Then, the normalised values for the SE and EE, denoted as SE_{norm} and EE_{norm} , respectively, are obtained as follows:

$$\text{SE}_{\text{norm}} = \frac{\text{SE}}{\text{SE}_{\max}} \quad (5)$$

$$\text{EE}_{\text{norm}} = \frac{\text{EE}}{\text{EE}_{\max}} \quad (6)$$

where $\{\text{SE}_{\text{norm}}, \text{EE}_{\text{norm}}\} \in [0, 1]$. The result of the normalisation process is illustrated in Fig. 2-a) for the SE and the EE (on the left), and their corresponding scatter plot representation (on the right). As shown in Fig. 2-a), the normalised values generally exhibit a narrow distribution around low values, since the maximum achievable values can be significantly higher than those observed. Additionally, SE_{norm} and EE_{norm} do not directly facilitate the interpretation of the resulting scatter plot. To address this, the proposed solution involves dividing SE_{norm} and EE_{norm} by predefined reference values, followed by applying a log transformation. This approach enhances the interpretability of the scatter plot by highlighting values above and below the selected reference. The choice of reference values can be based on either empirical data or industry benchmarks. An industry benchmark is often more appropriate, enabling fair comparisons between BSs from different MNOs or across different periods. In this study, the mean values of SE_{norm} and EE_{norm} were chosen as the reference values, SE_{ref} and EE_{ref} , respectively. This selection ensures a balanced representation across all categories, positioning the scatter plot's centre at the expected values for both metrics, thus facilitating relative comparisons. The newly normalised values, denoted as z_{SE} and z_{EE} , are defined as follows:

$$z_{\text{SE}} = \log\left(\frac{\text{SE}_{\text{norm}}}{\text{SE}_{\text{ref}}}\right) \quad (7)$$

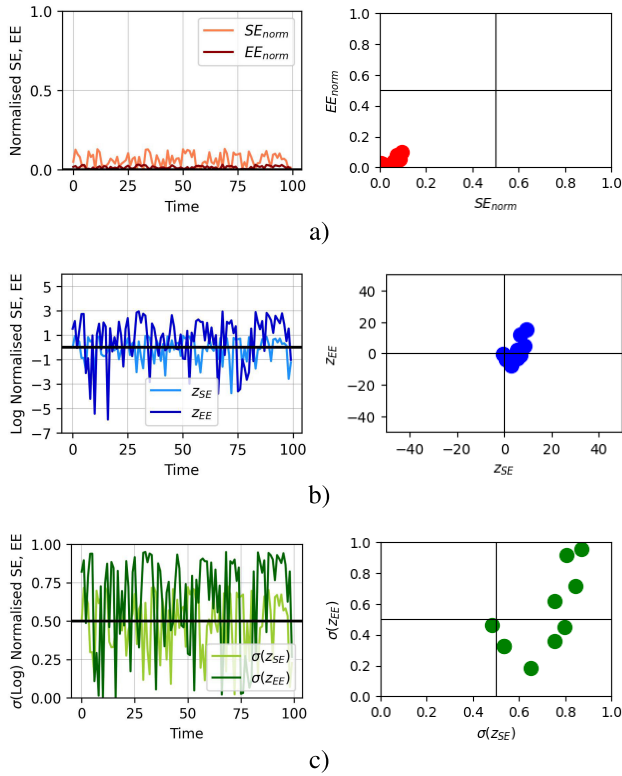


FIGURE 2. Example of the applied transformations to the original SE and EE values (on the left) and the corresponding scatter plot representation (on the right): a) Normalisation considering maximum values per RAT; b) Log transformation of the normalised values; c) Sigmoid transformation of the log-normalised values.

$$z_{EE} = \log\left(\frac{EE_{norm}}{EE_{ref}}\right) \quad (8)$$

The result is exemplified in Fig. 2-b), where the aim is to assign a positive output, $z_{SE} > 0$, when $SE_{norm} > SE_{ref}$, and $z_{EE} > 0$, when $EE_{norm} > EE_{ref}$; otherwise, negative values are assigned. The split of the z_{SE} and z_{EE} values around zero allows for a scatter plot representation where the interplay between these two metrics can be segmented into four classes. Although this representation enables a relative comparison between the z_{SE} and z_{EE} , it presents an unbounded scale which can hamper a clear graphical representation and the corresponding class definitions. To constrain the dynamic range of the transformed SE and EE values to a fixed range between 0 and 1, a sigmoid function was applied according to:

$$\sigma(z_{SE}) = \frac{1}{1 + e^{-z_{SE}}} \quad (9)$$

$$\sigma(z_{EE}) = \frac{1}{1 + e^{-z_{EE}}} \quad (10)$$

Among other normalisation functions, such as the ReLU [39], [40] or the hyperbolic tangent, the sigmoid function was selected due to its unique properties, particularly its monotonicity, which preserves the relative ordering of values, and its sensitivity to extreme inputs, enabling a clear

distinction between high and low performances. Moreover, as highlighted in [41], the sigmoid is widely used for normalising inputs, especially when a probability-like interpretation is needed. Additionally, as shown in Fig. 2-c), this transformation ensures that the centre of the scatter plot (coordinates: x-axis = 0.5, y-axis = 0.5) aligns with the predefined reference values.

IV. ENERGY SUSTAINABILITY FRAMEWORK

This section presents the ESF. Leveraging the data obtained from a live network containing observed PM and Energy Management (EM) indicators, the framework extends beyond the ESC's deterministic computation, which can be derived using the methodology described in the preceding section, by identifying root cause factors influencing the ESC. To achieve this, a supervised ML model was developed to predict the ESC, using the deterministic computed values as ground truth for training while excluding indicators with close dependencies on those values. This approach explores indirect relationships between the ESC and other observed indicators, providing insights for potential optimisation strategies.

A. IMPLEMENTATION STEPS

As illustrated in Fig. 3, the framework is structured in five sequential steps:

1) STEP 1 - DATA COLLECTION

The first step involves characterising the performance and energy consumption of each BS. This requires collecting and processing all available 4G and 5G PM and EM indicators from the considered RAN suppliers. Network topology information was also extracted for the considered BSs, including frequency bands, duplex mode, antenna models, and others. The description of each indicator is defined by the respective RAN supplier, following the standardised references in [42], [43], and [44]. This step is detailed in Section V.

2) STEP 2 - ESC COMPUTATION

This step, described in Section VI, involves the computation of the ESC for 4G, 5G and 4G/5G, aggregated per location, as defined in the preceding section. This computation serves as the ground truth for training the ML model in the next step.

3) STEP 3 - ML MODEL AND FEATURES IMPORTANCE

This step, detailed in Section VII, focuses on developing a supervised ML model that uses the previously computed deterministic ESC values as the target for training. The model incorporates all indicators gathered in step 1 while discarding, based on domain expertise, those with evident relationships to the deterministic computation (e.g., Physical Resource Block (PRB) usage vs Traffic). The resulting indirect indicators denote potential root cause links for the ESC. The importance of each indicator (model feature) in predicting the ESC

(model outcome) was quantified using Shapley values [45]. Finally, the magnitude of these values was used to rank the observed indicators according to their impact on the model outcome.

4) STEP 4 - MODEL EVALUATION AND FEATURE REDUCTION

The fourth step pertains to Section VIII, focusing on assessing the model's robustness in predicting the ESC relying on metrics like the F1-Score while minimising the number of used indicators. This process involves aggregating Shapley Values, accounting for the actual classes predicted by the model.

5) STEP 5 - ROOT CAUSE ANALYSIS

Building on the variables identified in step 4, the final step, detailed in Section IX, aims to identify optimisation approaches by analysing how these variables interact with the ESC. This qualitative assessment relies on expert knowledge to establish plausible causal relationships between the input variables and their root cause impact on the BSs' classification.

B. ESF MATHEMATICAL SUMMARY

A summary of the main notations used throughout this paper is provided in Table 1. The ESF considers a set of BSs and the corresponding PM data at each observation, $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$, where $\mathbf{x}_n \in \mathbb{R}^P$ are the indicators associated with the n^{th} observation, with P indicating the total number of indicators, and N the number of observations. At each observation, the ESC is computed using the formulation introduced in Section III, resulting in the vector of classes $\mathbf{Y}$. This vector serves as the model labels and is subsequently encoded using one-hot encoding [46], resulting in $\mathbf{Y}^o$, a binary representation of the categorical data. For instance, if $y_n = A$, its one-hot encoded representation would be the vector $\mathbf{y}_n^o = [1, 0, 0, 0]$. This encoding allows ML models to process categorical variables without assuming any ordinal relationships, thereby improving the effectiveness of model training. Additionally, a supervised ML model is trained to predict the classes, $\hat{\mathbf{Y}}$, based on $\mathbf{X}$, by adjusting

TABLE 1. Main notations used in this paper.

Notations	Descriptions
$c \in \{A, B, C, D\}$	The energy sustainability classes
N	The total number of observations in the dataset
P	The number of indicators in the dataset
M	The number of indicators after feature selection
$\mathbf{X} \in \mathbb{R}^{N \times P}$	The indicators dataset across N observations
$\mathbf{x}_n \in \mathbb{R}^P$	The indicators of the n^{th} observation
$\mathbf{Z} \in \mathbb{R}^{N \times M}$	The selected Z indicators after feature selection
$\mathbf{Y} \in \{A, B, C, D\}^N$	The calculated classes
$y_n \in \{A, B, C, D\}$	The class for the n^{th} observation
$\hat{\mathbf{Y}} \in \{A, B, C, D\}^N$	Predicted classes using a ML model
$\hat{y}_n \in \{A, B, C, D\}$	Predicted class for the n^{th} observation
$\mathbf{Y}^o \in \{0, 1\}^{N \times 4}$	One-hot encoding of classes
$\mathbf{y}_n^o \in \{0, 1\}^4$	One-hot encoding of the n^{th} class observation
$\mathbf{W} \in \mathbb{R}^{4 \times P}$	The parameters of the ML model
$\mathbf{w}_k \in \mathbb{R}^P$	The parameters for the k^{th} class
$\mathbf{S} \in \mathbb{R}^{N \times P \times 4}$	Shapley values per feature and per class
$\mathbf{S}' \in \mathbb{R}^{N \times P}$	Shapley values for the predicted class
$\bar{\mathbf{S}}' \in \mathbb{R}^P$	Average Shapley values for the predicted class
$\bar{s}'_p$	Average Shapley value for the p^{th} feature

its parameters $\mathbf{W}$. This model aims to identify potential root-cause factors for $\mathbf{Y}$. Subsequently, a feature selection process is implemented to create a reduced set containing M indicators (instead of P) resulting in a shorter indicators dataset, $\mathbf{Z}$ (instead of $\mathbf{X}$). To obtain the reduced feature set, Shapley values, $\mathbf{S}$, were calculated to assess the importance of each indicator $p \in P$ in the predicted classes, $\hat{\mathbf{Y}}$. Ultimately, the reduced feature set is derived by calculating the Shapley values for the actual predicted class, $\mathbf{S}'$, and sorting the model features in descending order based on the corresponding absolute average Shapley value for the p^{th} feature, $|\bar{s}'_p|$. This reduced feature set comprises the most influential features impacting the ESC, serving as key indicators for targeted optimisation actions derived from root cause analysis. Algorithm 1 outlines the complete five-step ESF procedure:

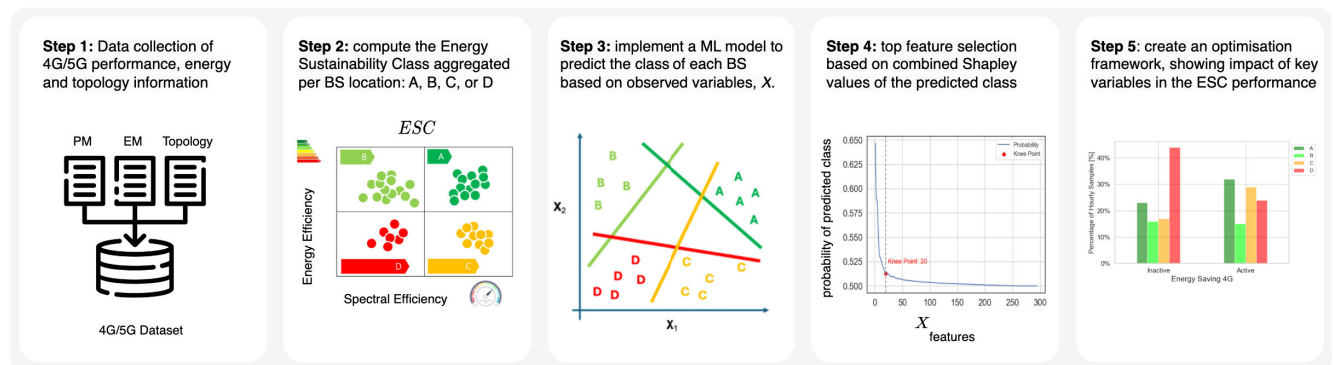


FIGURE 3. Five-step framework for ESC optimisation in live mobile telecommunications networks.

Algorithm 1 Five Step Energy Sustainability Framework.

- 1) **Data Collection and Preprocessing**
 - a) Extract BS data and compute synthetic features.
 - b) Compute $\mathbf{X} \in \mathbb{R}^{N \times P}$, comprising all BS' features.
- 2) For $n = 1$ to N : do **ESC Computation**
 - a) Compute the ESC : $\mathbf{y}_n \in \{A, B, C, D\}$.
 - b) Apply one-hot encoding to $\mathbf{y}_n$, yielding $\mathbf{y}_n^o$.
- 3) **Feature Importance Estimation**
 - a) Train a ML model to compute $P(\mathbf{Y}^o | \mathbf{X}, \mathbf{W})$.
 - b) Compute $\mathbf{X}, \mathbf{W} \rightarrow \hat{\mathbf{Y}}$ using the ML model.
 - c) Compute Shapley values per feature & class $\hat{\mathbf{Y}} \rightarrow \mathbf{S}$.
 - d) Compute average Shapley values per feature $\mathbf{S} \rightarrow \bar{s}_p$.
- 4) **Feature Reduction**
 - a) Rank the P features based on their $|\bar{s}_p|$.
 - b) Compute M using an elbow detection algorithm.
 - c) Select top M features to create $\mathbf{Z}$, where $M < P$.
- 5) **Output: Root Cause Factors**
 - a) $\mathbf{Z} \rightarrow \hat{\mathbf{Y}}$ indicate ESC potential root cause factors.

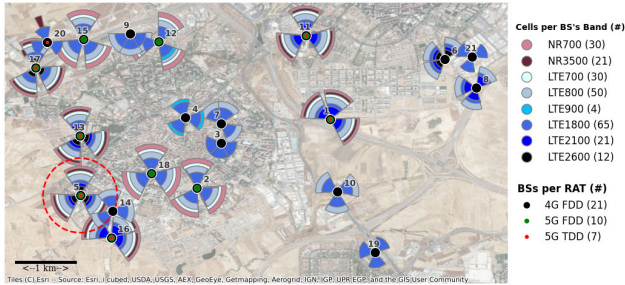


FIGURE 4. Geographical distribution of the BSs, with a colour scheme distinguishing frequency bands and technologies.

V. DATA COLLECTION

This section pertains to the first step of the ESF, starting by presenting the data used in the experimental analysis, which was obtained from live 4G and 5G BSs. After this, a brief data exploration is performed on some key indicators, such as traffic and energy consumption, to assess the current maturity level of these networks.

A. DATA DESCRIPTION

The data used in this work was extracted from a universe of 233 cells belonging to 38 BSs, deployed across 21 adjacent locations, spanning an area of roughly 35 km² in an urban/suburban area. The considered BSs are depicted in Fig. 4, with each location identified by a unique reference number, where each cell is represented according to its frequency band and each BS according to its RAT/duplex mode.

The frequency bands are arranged based on their theoretical coverage range: lower frequencies (e.g., 700, 800, 900 MHz) are positioned on the periphery of the BS

TABLE 2. Characterisation of the considered BSs.

RAT	Mode	BSs [#]	Shared BSs [#]	Cells [#]	avgCells per BS	avgBW per BS [MHz]	Bands [MHz]
4G	FDD	21	12	182	8.7	246.6	700, 800, 900, 1800, 2100, 2600
5G	FDD	10	10	30	3.0	60.0	700
	TDD	7	6	21	3.0	270.0	3500

symbol to indicate broader coverage, while higher frequencies (e.g., 1800, 2100, 2600, 3500 MHz) are nearer the centre, indicating shorter coverage. Moreover, the prefixes “LTE” and “NR” exhibited in the figure’s legend denote respectively the coverage bands used in 4G and 5G. For every considered BS, PM and EM indicators were collected, characterising their performance and energy consumption over N observations. Overall, 719 distinct indicators were collected: 464 on 4G, and 255 on 5G BSs, respectively. The observation corresponds to a continuous 37-day period, with hourly granularity, exclusively from a single network equipment provider.

Complementary topology information was obtained for the considered BSs, including frequency bands, antenna models, geographical coordinates, duplex modes, RAN sharing and others. Table 2, provides a detailed characterisation of these BSs covering aspects like RAN sharing with other MNOs, the average number of cells, the average bandwidth for downlink and uplink, and the duplex mode configurations. In the FDD configurations used, the available spectrum is divided into independent and symmetric downlink and uplink bands. In contrast, TDD configurations use the same spectrum for both downlink and uplink, typically allocating around 70% for downlink, with the remaining 30% dynamically adjustable. The considered BSs include 21 operating in 4G, all in FDD mode, and 17 in 5G, with the latter including seven in TDD mode and ten in FDD mode - 6 of which are collocated with TDD. In total, the 4G BSs are deployed across 21 distinct locations, while the 5G BSs cover 11 distinct locations. The scale of the installed capacity is emphasised by the high average bandwidth per BS location, reaching up to 500 MHz when considering the combined bandwidth of both 4G and 5G RATs. This is visually exemplified in Fig 4, focusing on location number 5, enclosed by a red dotted circle.

B. DATA EXPLORATION

To complement the initial data description, an additional analysis was performed shedding light on key operating aspects of this network. Fig. 5 depicts the CDF for three indicators involving capacity demand, coverage/interference level, and energy consumption. The first indicator is the daily average traffic per cell, both in the downlink and uplink directions, comprising 4G and 5G, as shown in Fig. 5-a). The second indicator - average Channel Quality Index (CQI) - measures the average downlink interference levels measured per cell by UEs and reported to the respective

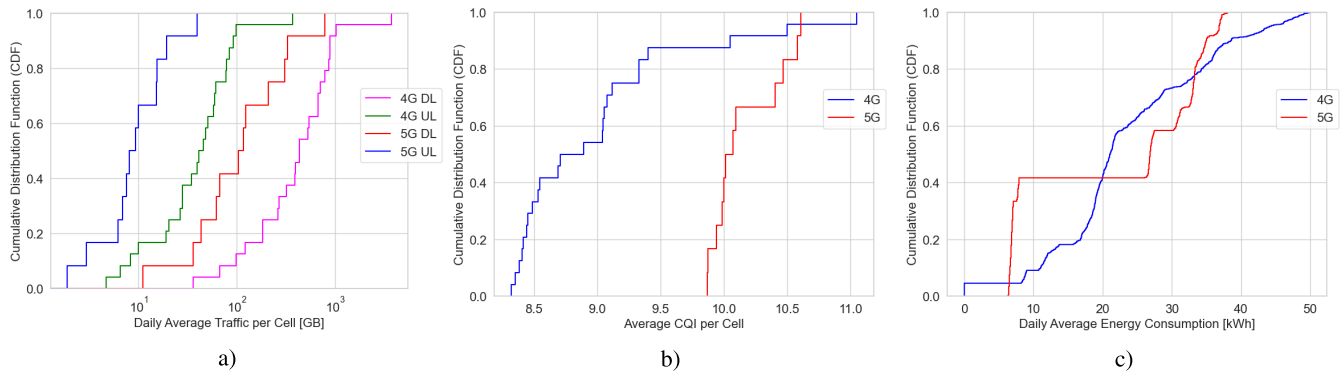


FIGURE 5. Data exploration of three main KPIs: a) Cumulative Distribution Function (CDF) of daily average traffic per cell and per RAT; b) CDF of average CQI per BS and per RAT; c) CDF of daily average energy consumption per BS and per RAT.

BS. A lower CQI indicates higher interference, while a higher CQI suggests less interference. For 5G, this measure combines data from both FDD and TDD modes, as shown in Fig. 5-b). The third indicator, depicted in Fig. 5-c), measures the daily average energy consumption per BS and per RAT (the smallest granularity available). From this exploratory data observation, several insights emerge:

- **Downlink vs. Uplink Traffic** - Downlink traffic is significantly higher than uplink traffic, exceeding it by approximately ten times in both RATs, as shown in Fig. 5-a). Given the dominance of downlink traffic, and for simplification purposes, the SE and EE metrics in this paper were calculated based solely on the downlink direction. However, the methodology can be easily extended to include downlink and uplink directions.
- **4G vs 5G Traffic Load** - As depicted in Fig. 5-a), the median cell traffic load in 4G surpasses that in 5G by approximately four-fold (432 GB/day vs 103 GB/day). This reinforces that the considered 5G network is in its early adoption stages, with underutilised capacity.
- **4G vs 5G Radio Channel Quality** - The 4G cells present a wide distribution of interference levels. In contrast, 5G exhibits a narrower distribution concentrated at low interference levels (high CQI), as depicted in Fig. 5-b). These asymmetric distributions denote the higher maturity level of the 4G network. Conversely, the 5G distribution shows that the MNO's current strategy targets the 5G deployment only at selected locations.
- **4G vs 5G Energy Consumption** - Despite of the fact that 4G presents a higher median traffic per cell compared to 5G, the corresponding median energy consumption per BS is higher in 5G, as shown in Fig. 5-c).

In summary, the data exploration analysis reveals distinct maturity stages in the 4G and 5G networks. While all BSs have 4G, the 5G BSs are limited to approximately 50% of the available BS locations

VI. ENERGY SUSTAINABILITY CLASS COMPUTATION

As outlined in Section IV, the second step in the ESF involves computing the aggregated ESC for each location

and the considered RATs, based on the corresponding BSs. The primary objective is to create a relative performance assessment of the BSs' locations, considering the SE and EE domains. Moreover, the ESC can be computed through three distinct scenarios, described as follows:

- **4G** - In this scenario, the class calculation exclusively takes into account the 4G BSs along with their corresponding PM and EM indicators.
- **5G** - In this scenario, only the 5G BSs and the respective PM and EM indicators are considered.
- **4G/5G** - In this scenario, the class calculation aggregates the contributions of both 4G and 5G BSs, with their weights determined by the downlink traffic volume.

Fig. 6 illustrates the ESC scatter plot representation for each scenario, after applying the ESF to the considered BSs. In these plots, each circle represents a BS location, aggregating a specific RAT or a group of RATs. The coordinates of the circles are the average value over each dimension, computed for each BS location and considering the entire observation period. Moreover, the colour of each circle represents the ESC, while the number associated with each circle is the unique location identification introduced in Fig. 4. It is important to highlight that the centre of each scatter plot, in every considered scenario, represents the average value of the SE and EE. In Fig. 7 the same information is provided through a geographical representation of the ESC across the three scenarios. By collectively examining Figs. 4, 5, 6, and 7, several insights emerge regarding the results in each considered scenario:

- **4G** - As depicted in Fig. 6-a), there is a heterogeneous distribution of the ESC across the different the BSs' locations with 4G.
- **5G** - The observation of Fig. 6-b) reveals no 5G BSs' locations classified in class A. Moreover, in 7 out of the 11 BSs locations with 5G, the classification falls within classes C and D, which are characterised by the under-average EE, denoting a low ratio of traffic per energy consumption. Further inspection of Fig. 4 reveals that the BS's location with lower EE exclusively operates in 5G TDD mode, with 270 MHz

of installed bandwidth. Conversely, the BSs' locations with higher EE exclusively use the 5G FDD mode, constrained to 60 MHz of bandwidth per BS's location. This observation reinforces that the 5G network is in its early adoption phase, resulting in underutilised capacity, particularly in BSs using TDD mode, leading to sub-optimal efficiency both in SE and EE.

- **4G/5G** - Considering the perceived low utilization in 5G BSs and the lower number of BSs than in 4G, the aggregated 4G/5G ESC computation in Fig. 6-c) aligns with the outcome in Fig. 6-a), where only 4G BSs were considered. However, in specific cases, such as the one illustrated by BS with reference number 5, the aggregated classification falls in class C, while in the 4G perspective, this BS was classified in class A. This underscores that the contribution of 5G exerts a negative impact on the aggregated ESC for this BS location.

VII. ML MODEL AND FEATURES IMPORTANCE

This section pertains to step 3 in the ESF, as presented in Section IV (see Fig. 3). In this step, the data collected in the previous phase is utilised to train a supervised ML model to predict the ESC. The goal is to identify the most influential factors associated with the classifications.

A. DATA PREPARATION

The supervised ML model requires a curated dataset, where PM and EM features that are directly or indirectly involved in

the ESC calculation, notably traffic and energy consumption data, are removed. The data engineering process includes eliminating the undesired entries and creating additional synthetic features using raw dataset metrics (*e.g.*, average CQI per BS). Table 3 summarises the dataset size, the number of removed and added features, and the resulting set of features for the ML model.

TABLE 3. Dataset preparation for the ESF ML analysis.

Aggregation	Base Input Features [#]	Removed Features [#]	Added Features [#]	Features for Curated ML Model [#]
4G	464	174	8	298
5G	255	140	7	122
4G/5G	719	314	14	419

- 1) **UeCqiAvg** (2) - Computes the average of the individual CQI measurements reported by the UEs to the BS (ranging from 0 to 15). Higher values indicate better radio channel quality and less interference, as detailed in Section V-B.
- 2) **McsDLAvg** (2) - A synthetic feature computed for both 4G and 5G, representing the average MCS applied in the downlink direction. The MCS range is between 0 and 31, where higher values correspond to higher-order modulation schemes (higher throughput).
- 3) **McsULAvg** (2) - Similar to the previous synthetic feature, but for the uplink direction.

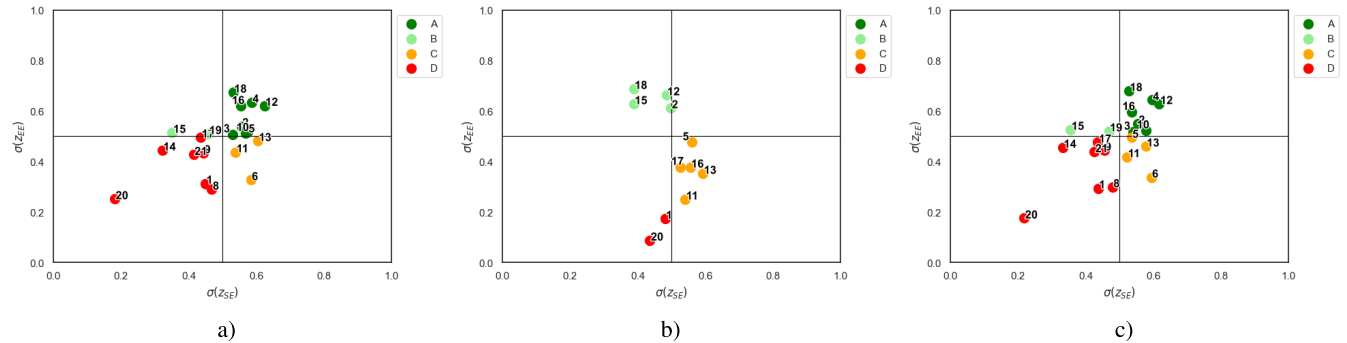


FIGURE 6. Scatter plot representation for the ESC computation aggregated per BS location: a) considering only the 4G BSs; b) considering only the 5G BSs; c) considering the aggregated contribution of 4G and 5G BSs.

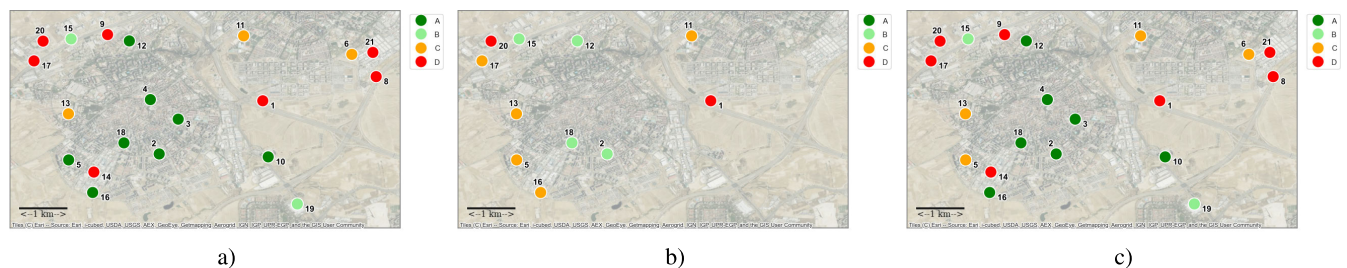


FIGURE 7. ESC computation per BS location and by geographical distribution: a) considering only the 4G BSs; b) considering only the 5G BSs; c) considering the aggregated contribution of 4G and 5G BSs.

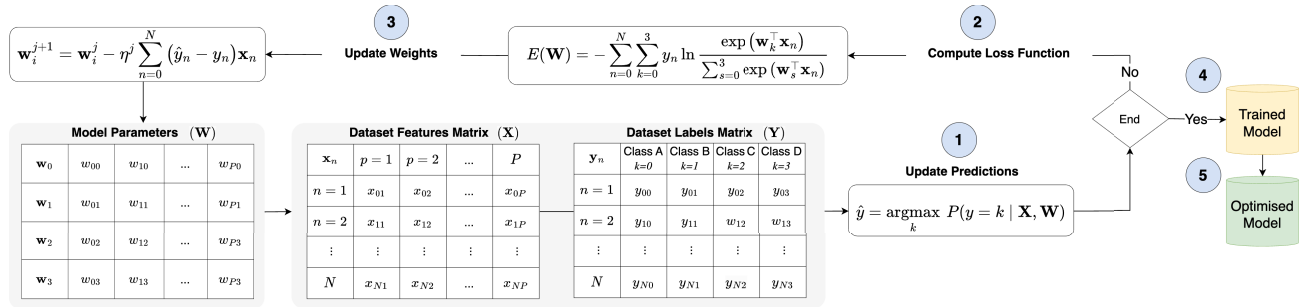


FIGURE 8. Implementation process of the ESF ML classification model.

- 4) **Avg_UL_RSSI** (1) - Computed only for the 4G scenario, representing the average Received Signal Strength Indication (RSSI) in the uplink direction (in dBm), derived from the RSSI values reported by all UEs to the BS and averaged across all receiving antennas.
- 5) **Cells_Sharing_Ratio** (2) - Derived from topology data and calculated for 4G and 5G, quantifies the extent of shared cells with another MNO. It ranges from 0 to 1, with 1 indicating no sharing and 0 indicating complete sharing with another operator.
- 6) **N_Cells** (2) - A synthetic feature derived from topology data that counts the number of cells per BS in both 4G and 5G. This feature provides information about the number of overlaid cells, including those operating on different frequency bands.
- 7) **Distance_2** (2) - Derived from topology data, measures the average distance to the two nearest BSs, which helps to identify potential coverage gaps or interference zones.
- 8) **5G_BW** (1) - Derived from topology data, denotes the total 5G spectrum at each 4G location (excluded in the combined 4G/5G scenario since it influences the ESC).
- 9) **Duplex_Mode** (1) - Derived from topology data, representing the operation mode of collocated 5G BSs. Possible values are 0 (FDD), 1 (TDD), and 2 (FDD/TDD).

In total, 15 additional synthetic features were included across the different scenarios: 8 for 4G and 7 for 5G. However, in the combined 4G/5G scenario, only 14 features were used, as shown in Table 3, since the 5G_BW feature was excluded. This exclusion was necessary because it directly influences the ESC calculation in the combined scenario.

B. ESF ML MODEL IMPLEMENTATION

The ML model complements the deterministic computation of the ESC, as described in Section III, by identifying key factors influencing the classification. Implemented as a multi-class classification problem with four categories (A, B, C, and D), it leverages the available network data to determine which variables most impact each class.

The decision was to perform the ML implementation using the Light Gradient-Boosting Machine (LGBM) [47], due to its ability to capture non-linear relationships and interactions

between variables, through sequential decision trees that iteratively correct errors. However, the framework is flexible and allows for model complexity to be increased as needed. The ESF ML model implementation process comprises five steps as illustrated in Fig. 8, which are detailed as follows:

1) CLASS PREDICTIONS

In the first step, 80% of the dataset observations are used for training involving, for each observation, the computation of which of the classes yields the highest probability taking into account the current values of the model parameters, $\mathbf{W}$ and the features, $\mathbf{X}$. The normalised probabilities for each class, c , are computed using the softmax function [48], given by:

$$P(y = c | \mathbf{X}, \mathbf{W}) = \frac{\exp(\mathbf{w}_c^\top \mathbf{X})}{\sum_{k=0}^3 \exp(\mathbf{w}_k^\top \mathbf{X})}. \quad (11)$$

where $\mathbf{w}_c$ denotes the model weight parameters for the selected class, c , and $\mathbf{w}_k$ represents the weight parameters for each k^{th} class. The predicted class is then given by:

$$\hat{y} = \underset{k}{\operatorname{argmax}} P(y = k | \mathbf{X}, \mathbf{W}). \quad (12)$$

2) LOSS FUNCTION COMPUTATION

The objective of the second step is minimising the prediction error using the negative Log-Likelihood (nLL) function. This involves maximising the Maximum Likelihood Estimation (MLE) function, denoting the joint probability of observing $\mathbf{Y}$, given $\mathbf{X}$ and $\mathbf{W}$, described in [49] by:

$$P(\mathbf{Y} | \mathbf{X}, \mathbf{W}) = \prod_{n=0}^N \prod_{k=0}^3 P(y_n = k | \mathbf{x}_n, \mathbf{w}_k)^{y_{nk}} \quad (13)$$

where $P(y_n = k | \mathbf{x}_n, \mathbf{w}_k)$ is the probability of observing the class k for the n^{th} observation, given its features $\mathbf{x}_n$ and the weight parameters $\mathbf{w}_k$. The vector y_n has four binary components, one for each class. Additionally, y_{nk} is a binary component of the vector y_n using the one-hot encoding approach. Finally, the loss function is computed by taking the derivative of (13) and finding the values for which it equals 0. As shown in [49], this is equivalent to minimising the nLL,

resulting in the desired loss function:

$$E(\mathbf{W}) = - \sum_{n=0}^N \sum_{k=0}^3 y_n \ln \frac{\exp(\mathbf{w}_k^\top \mathbf{x}_n)}{\sum_{s=0}^3 \exp(\mathbf{w}_s^\top \mathbf{x}_n)} \quad (14)$$

3) WEIGHTS UPDATING

The third step involves computing the gradient of (14) and determining the weight values that minimise the nLL loss function. Since this equation does not have a closed form, an optimisation algorithm is required to find a solution. In the ESF ML model implementation, the choice for this optimiser relied on the Stochastic Gradient Descent (SGD) [50]. This algorithm has the advantage of reducing the computational cost, with faster iterations, at the expense of a lower convergence rate. This is accomplished by replacing the actual gradient, calculated from the entire data set, with an estimate based on a randomly selected data subset. During the training cycles, at each iteration j , the model parameters are adjusted to minimise the loss function, as shown in:

$$\mathbf{w}_i^{j+1} = \mathbf{w}_i^j - \eta^j \nabla_{\mathbf{w}_i} E, \quad (15)$$

where η is a model hyperparameter called the learning rate.

4) TRAINED MODEL

The fourth step is completed when the termination condition is met, after iteratively repeating steps 1 to 3 using 80% of the dataset. In implementing the ESF ML, early stopping was used to halt training if the model's performance showed no improvements after 100 iterations. Once the termination condition is satisfied, the model parameters, $\mathbf{W}$, are stored.

5) OPTIMISATION AND REGULARISATION

The final step entails the optimisation of the ESF ML model hyperparameters. In this procedure, different combinations of the available hyperparameters are tested, repeating steps 1 to 4, while evaluating the achieved results using 10% of the dataset as the validation set. The goal of the optimisation process is tuning each hyperparameter while balancing model complexity and over-fitting with generalisation. Finally, the model training is executed with the best hyperparameters configuration and the corresponding parameters, $\mathbf{W}$, are stored. Considering the utilisation of the LGBM model the corresponding available hyperparameters can be summarised in two groups: optimisation and regularisation, comprising eight hyperparameters, described as follows:

1) Optimisation

- num_leaves**: the maximum number of leaves per tree.
- n_estimators**: the number of boosting iterations or trees in the model.
- min_child_samples**: the minimum samples per leaf.
- bagging_fraction**: the fraction of data to be used for each boosting iteration.
- max_depth**: the maximum depth of each tree.
- learning_rate**: the rate at which model parameters are updated, as described in (15).

2) Regularisation

- lambda_l1**: the L1 regularisation term on model parameters, also known as Lasso regularisation.
- lambda_l2**: the L2 regularisation term on model parameters, also known as Ridge regularisation.

Although a detailed explanation of each hyperparameter is beyond the scope of this paper, it is important to address the regularisation terms due to their impact on model interpretability. Regularisation differentiates feature importance by adjusting parameter magnitudes. L1 (Lasso Regularisation) induces sparsity by forcing some parameters to zero, while L2 (Ridge Regularisation) reduces parameter magnitudes without necessarily eliminating them. Both methods mitigate over-fitting by adding a penalty term to the loss function. The L1 regularisation expression is shown in (16), where the penalty term is the product of the hyperparameter λ and the L1 norm of the weights, which is the sum of their absolute values. A similar expression applies to L2 regularisation, with the squared magnitude of the weights used in the penalty term.

$$\hat{\mathbf{w}}_{\text{sparse}} = \arg \min_{\mathbf{w}} E(\mathbf{W}) + \lambda \|\mathbf{W}\|_1 \quad (16)$$

The optimisation procedure to identify the best hyperparameter configuration considered 500 trials per scenario. For this purpose, optimisation software was used, where the choice was the Optuna framework [51]. This choice relied on Optuna being easily adapted to complex code, supporting test parallelization and pruning of unsuccessful tests. Table 4 exhibits the range of tested values for each hyperparameter, and the final selected values per scenario.

TABLE 4. Best hyperparameter settings per scenario.

Hyperparameters	Range		Selected Value		
	Min	Max	4G	5G	4G/5G
num_leaves	5	123	110	85	45
n_estimators	20	160	160	130	135
min_child_samples	5	100	40	50	35
bagging_fraction	0.5	1.0	0.90	0.94	0.54
max_depth	2	20	14	16	16
learning_rate	1e-5	0.5	0.09	0.30	0.16
lambda_l1	1e-3	20	0.32	3.93	0.39
lambda_l2	1e-3	20	16.44	3.34	15.39

C. MODEL EVALUATION

For evaluation purposes, the reserved test set was employed, comprising 10% of the total dataset. The performance evaluation for each class is assessed independently, treating the problem as binary. This binary classification output is represented by a 2×2 confusion matrix. From these matrices, four fundamental measures are derived: True Positives (TPs), True Negatives (TNs), False Positives (FPs), and False Negatives (FNs). Subsequently, four performance metrics, commonly used in binary classification problems, can be computed:

- **Accuracy:** The ratio of correctly classified examples to the total number of examples.

$$\text{Accuracy} = \frac{TN + TP}{TN + FP + TP + FN} \quad (17)$$

- **Precision:** The percentage of correctly classified positive examples in a class among the instances predicted.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (18)$$

- **Recall:** The percentage of positive examples correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (19)$$

- **F1-Score:** The harmonic mean between Precision and Recall.

$$F1\text{-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (20)$$

In the context of multi-class classification, the model's overall performance can be gauged by the weighted average of the Precision, Recall and F1-Score, considering the number of examples classified in each category. The model evaluation results are presented in Tables 5-a), 5-b), and 5-c), using the previously introduced performance metrics applied to the test set. The examination of the previous results highlights the following insights and comments:

- **Absolute Analysis** - The ML models demonstrated robust predictive capabilities for the ESC at each observation, achieving high-performance metrics across all considered scenarios. Notably, precision values ranged between 0.78 in the 5G scenario and 0.86 in the 4G/5G scenario.
- **Relative Analysis** - The ML model performed worst in the 5G scenario and best in the combined 4G/5G scenario. The former supports the preliminary findings from data exploration, which indicated that the 5G network had fewer indicators and due to a lesser number of BSs compared to 4G. The latter confirms that the proposed methodology is effective for multi-RAT BSs.

Notably, the ESC of a given BS is not constant over time, exhibiting dynamic behaviour resulting from the diverse traffic profiles of the underlying cells. These hourly readings contrast the aggregated perspective, where the figures at each observation are averaged over the full dataset

period. The impact of these two different time aggregation approaches in the data representation is evident in Table 5-b), where 68 examples were observed in class "A" in the 5G scenario, despite the respective time aggregated scatter plot representation showing no BSs in this class, as presented in Fig. 6-b).

D. MODEL EXPLAINABILITY

The interpretability of the ML model plays a crucial role in the objective of the ESF. Hence, the aim is the identification of the most influential features impacting the model predictions and quantifying their respective contributions. This information will be leveraged for feature selection of the ML model, reducing complexity and enhancing interpretability.

1) COMPUTING FEATURE IMPORTANCE

In linear models like Logistic Regression (LR), the sign and magnitude of the model parameters, $\mathbf{W}$, directly relate to the probability of $\hat{\mathbf{Y}}$. In contrast, non-linear models such as LGBM lack this direct interpretability. Hence, to estimate the average contribution of specific features on predictions while remaining agnostic to the ML model, a model-agnostic explainability module was integrated into the ESF. This explainability module aims to estimate feature importance and indicate how variations in each feature affect predictions. An example of such a model-agnostic framework is provided by the "SHAP" Python library [52], which was chosen considering its robust results and simple implementation [53]. This library implements the calculation of the Shapley values [45] representing the significance of each model feature, at each observation.

When dealing with multi-class problems such as the ESC ML model, the obtained Shapley values can be interpreted as the difference between the predicted log odds and the average predicted log odds for each considered class. Hence, the log odds of predicting a specific class, c , increase with positive corresponding Shapley values and decrease otherwise. Consequently, as explained in [52] and [54], the probability of predicting c given $\mathbf{x}^p$ is determined as follows:

$$P(y = c \mid \mathbf{x}^p) = \frac{1}{1 + \exp(-|\bar{s}_{pc}|)} \quad (21)$$

TABLE 5. ESF ML model evaluation results per scenario.

(a) 4G.					(b) 5G.					(c) 4G/5G.				
Class	Precision	Recall	F1-Score	N_Obs. (#)	Class	Precision	Recall	F1-Score	N_Obs. (#)	Class	Precision	Recall	F1-Score	N_Obs. (#)
A	0.88	0.85	0.86	525	A	0.62	0.49	0.55	68	A	0.90	0.91	0.91	493
B	0.83	0.75	0.79	302	B	0.87	0.89	0.88	244	B	0.83	0.80	0.81	288
C	0.76	0.77	0.77	396	C	0.78	0.79	0.79	297	C	0.81	0.80	0.81	357
D	0.84	0.89	0.87	644	D	0.75	0.77	0.76	270	D	0.88	0.89	0.88	630
Total	0.83	0.83	0.83	1867	Total	0.78	0.79	0.78	879	Total	0.86	0.86	0.86	1768

where the term $|\bar{s}_{pc}|$ denotes the absolute value for the average of the Shapley values log odds for the p^{th} feature, x^p , in respect to class $c \in \{A, B, C, D\}$.

2) FEATURE IMPORTANCE USING SHAPLEY VALUES

The Shapley values computation leveraged the LGBM predictions to create $S \in \mathbb{R}^{N \times P \times 4}$ representing Shapley values over N observations, P features, and 4 classes. The Shapley values for each class, c , and for each feature, p , can be determined as the average absolute value computed across all observations, $|\bar{s}_p|_{c=0:3}$. Furthermore, the cumulative Shapley value of each feature can be determined as the sum of the average Shapley values for each class. This computation methodology is illustrated in Fig. 9-a), and the respective output is exemplified in Fig. 9-b), showcasing the computation of the top-10 cumulative Shapley values corresponding to each model feature for the 4G/5G scenario.

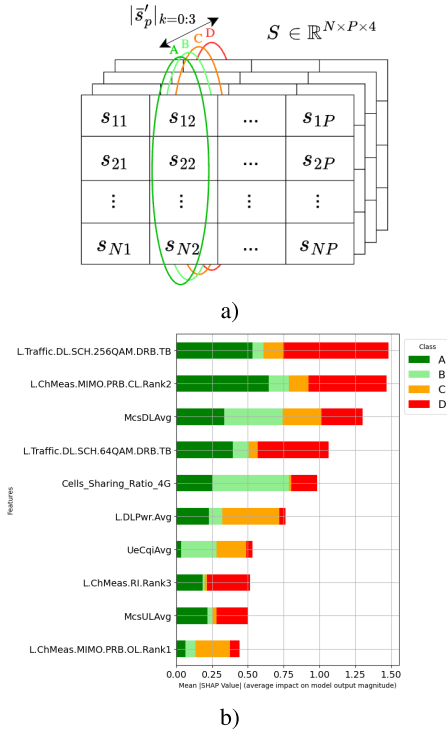


FIGURE 9. Computation of Shapley values per class: a) illustration of the calculation methodology; b) example for the 4G/5G scenario depicting the top-10 cumulative Shapley values.

VIII. FEATURE REDUCTION

This section represents the fourth step in the framework outlined in Section IV, focusing on reducing the feature sets to a minimal number of indicators while preserving the robustness of the LGBM model in predicting the ESC. The process involves the implementation of a methodology based on the average Shapley values to determine the most influential features to the model predictions, thereby reducing complexity and enhancing the model's interpretability. The

emphasis on interpretability is key for optimising the ESC, as discussed in Section IX.

A. SHAPLEY VALUES AGGREGATION

While the cumulative Shapley values computation facilitates understanding the interplay between each class and feature, it overlooks the actual occurrences of classes within the model predictions. Consequently, the resulting feature importance assessment might overemphasise features effective in predicting classes with low occurrence frequency, and conversely, underestimate those significant for classes with high frequency. A solution to address this issue is to compute the Shapley values for each feature while considering, at each observation, the predicted class based on the vector of predicted labels $\hat{Y} \in \mathbb{R}^{1 \times N}$, representing the model's predictions over N observations. Therefore, at each index n , the objective is to select the row of the tensor S corresponding to the predicted class $\hat{y}_n$. The final output is a matrix $S' \in \mathbb{R}^{N \times P}$, where each selected row is denoted as $S'_n \in \mathbb{R}^{1 \times P}$, and can be expressed as follows:

$$S'_n = S_{n,1:P,\hat{y}_n=k} \quad (22)$$

where, $S_{n,1:P,\hat{y}_n=k}$ denotes the row of the tensor S corresponding to the predicted class $\hat{y}_n = k$ for the observation n . Moreover, relying on the matrix S' , a vector, $\bar{S}' \in \mathbb{R}^{1 \times P}$, is created containing the average Shapley values characterising each feature across all observations, as depicted in Fig. 10.

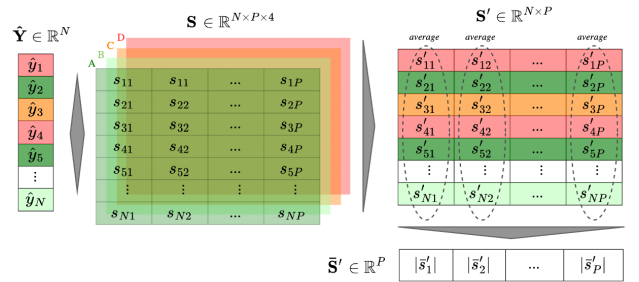


FIGURE 10. Shapley values aggregation process, with colours representing the predicted ESC, $\hat{Y}$, for different observations.

In the original representation, S , comprising Shapley values discriminated per class, the importance of each feature could only be established on a per-class basis, limiting the assessment of the impact of each feature. In contrast, the new Shapley values representation provided by the aggregation in S' , and subsequent computation of average values per feature is independent of the considered classes, although pondered by their relative prevalence in the model predictions. The benefit of this new representation is facilitating a direct assessment of the relative importance of each particular feature, which is more suitable for a feature selection process.

B. FEATURE SELECTION

As presented in Section VII, the ML model analysis involved three distinct scenarios (4G, 5G, and 4G/5G), each

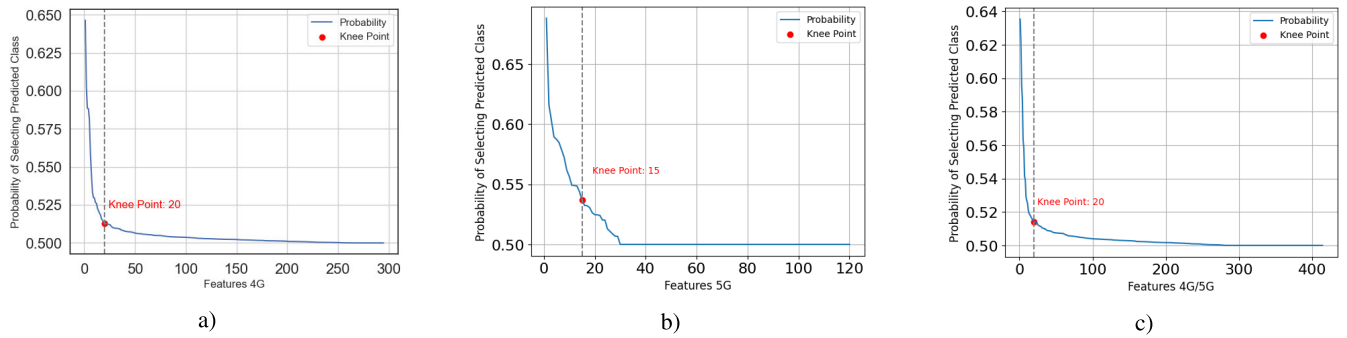


FIGURE 11. Feature selection criterion using knee point detection: a) 4G scenario; b) 5G scenario; c) 4G/5G combined scenario.

characterized by unique features and labels. Furthermore, these scenarios exhibit a substantial number of features: 295 in the 4G, 120 in the 5G, and 414 in the combined dataset scenario. Given the primary objective of identifying the most influential features for predicting the ESC, it becomes imperative to implement a feature reduction process. This process aims to eliminate redundant or non-informative variables while ensuring the resilience of the ML model's performance.

1) FEATURE RANKING

Building upon the methodology delineated in Fig. 10, the relative importance of the P features can be established by sorting them in descending order based on the magnitude of $|\bar{s}_p|$ for each $p \in P$. The resulting ranking denotes the effective relevance of each feature, considering the model's actual class predictions. Furthermore, this ranking can be refined by using (21) to convert the Shapley values into predicted class probabilities, conditioned on each feature p .

2) SELECTION CRITERION

A key aspect of feature selection is defining criteria for discarding features. In this analysis, scatter plot curves were created, with the x-axis representing the features, p , and the y-axis showing the probability of the model's predicted class conditioned on each feature. As illustrated in Fig. 11, these curves exhibit a convex decreasing shape that converges to a probability of 50%. This behaviour indicates that as feature importance diminishes, the model's predictive power approaches that of a random classifier. To determine the optimal selection point, a "knee point detection" algorithm was employed, identifying threshold values of 20, 15, and 20 features for the 4G, 5G, and 4G/5G scenarios, respectively. Based on the method outlined in [55], this algorithm identifies significant changes in curve behaviour by locating the point where the rate of change sharply increases.

While the curves for the 4G and 4G/5G scenarios fit well to a convex decreasing function, the 5G scenario displayed a unique "two-step shape," indicating that features beyond the 30th contribute little additional value. This behaviour is likely due to the early adoption phase of the 5G network,

characterized by low usage and limited variability in radio propagation scenarios. Nonetheless, the "knee-detection" algorithm effectively identified the curve's elbow point.

3) FINAL FEATURE SELECTION

Following the feature reduction process applied to the three scenarios, a concise list of 37 features emerged, comprising 21 indicators for 4G and 16 for 5G. An examination relying on domain knowledge of this list categorises these indicators into four groups: MIMO, Channel Quality, Energy Saving, and Network Topology. The visualisation in Fig. 12 illustrates the correspondence between each category type (on the left), the respective technology scenario (in the middle), and each characteristic feature (on the right), where the size of each feature's box is proportional to its importance in predicting the ESC. Table 6 provides a more detailed description of the reduced feature set, where the features are ordered following Fig. 12. It is noteworthy, that the four categories of indicators are associated with different aspects according to the functional context of each variable:

- **Channel Quality (22)** - This group includes 22 out of the 37 features in the reduced set. These features exert the highest impact on model prediction, especially in the 4G and 4G/5G scenarios. Key indicators in this category include MCSs, average power for downlink transmission, and the average CQI.
- **MIMO (7)** - This category includes 7 out of the reduced set of 37 features. The key features in this category encompass the number of independent MIMO spatial data streams.
- **Network topology (6)** - These features pertain to network topology variables such as the RAN sharing configuration, the number of cells per BS, and the inter-BS distance for 4G and 5G RATs.
- **Energy Saving (2)** - The energy-saving features comprise 2 out of the 37 features in the reduced set, exclusively associated with the 4G RAT.

4) MODEL PERFORMANCE WITH SELECTED FEATURES

After deriving the reduced feature sets for each scenario, a final validation is necessary to confirm the model

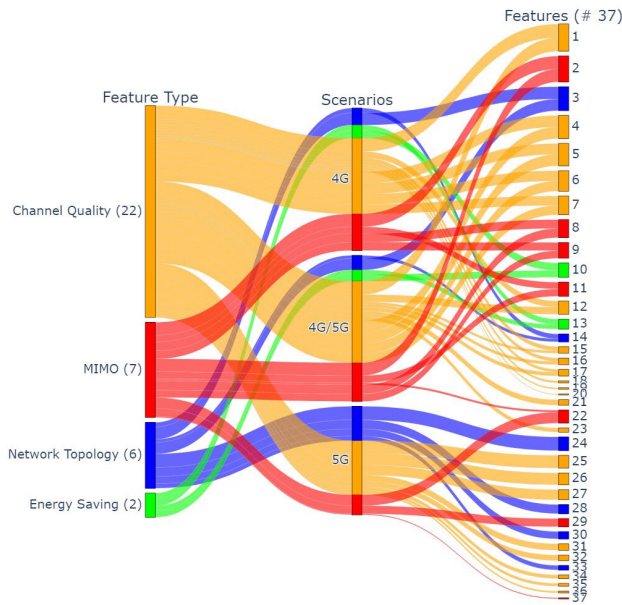


FIGURE 12. Reduced feature set per addressed scenario and feature type encompassing 37 4G and 5G indicators.

performance resilience while applying these reduced sets. To accomplish this, the ML models for each scenario were retrained and optimised using the methodology described in

Section VII. The final results were computed on the test set for each scenario, comparing the model performance with both the original and reduced feature sets.

The results are presented for the previously introduced performance metrics: Precision, Recall, and F1-Score. As depicted in Fig. 13, the model's performance remains consistent with the reduced feature sets across all scenarios. This validates the effectiveness of the feature reduction methodology applied to the original features. Furthermore, the results obtained from the ML model analysis and subsequent feature selection establish a basis for increased confidence in attributing any observed changes in outcomes to specific features, thereby reinforcing the robustness of the intended optimisation framework.

IX. ESC OPTIMISATION

This section delves into the ESC optimisation, as presented in step 5 of Fig. 3. This involves compiling a concise list of key variables grouped by their functional context and identifying their root-cause associations with the BSs' class. A qualitative assessment based on domain expertise explores the model results and presents an application example.

A. OPTIMISATION PROCESS

To recap the ESF and highlight the key actions, optimising the ESC of BSs' can be summarised in the following three stages:

TABLE 6. Reduced feature sets after applying a feature selection process relying on the aggregated Shapley values.

#	Features	RAT	Type	Description
1	L.Traffic.DL.SCH.256QAM.DRB.TB	4G	Channel Quality	The number of data blocks using 256QAM modulation scheme in downlink
2	L.ChMeas.MIMO.PR.B.CL.Rank2	4G	MIMO	The number of used downlink PRBs in closed-loop rank 2 mode
3	Cells_Sharing_Ratio_4G	4G	Network Topology	The percentage of 4G cells operated by the MNO within a BS location
4	L.Traffic.DL.SCH.64QAM.DRB.TB	4G	Channel Quality	The number of data blocks using 64QAM modulation scheme in downlink
5	McsDLAvg_4G	4G	Channel Quality	The average MCS index value used in 4G for scheduling in the downlink direction
6	L.DLPwr.Avg	4G	Channel Quality	The average downlink transmit power in a 4G cell
7	McsULAvg_4G	4G	Channel Quality	The average MCS index used in 4G for scheduling in the uplink direction
8	L.ChMeas.RI.Rank3	4G	MIMO	The number of times 3 independent streams are used in a 4G MIMO channel
9	L.ChMeas.MIMO.PR.B.OL.Rank1	4G	MIMO	The number of used downlink 4G PRBs in open-loop rank 2 mode
10	L.Cell.Unavail.Dur.EnergySaving	4G	Energy Saving	The duration of cell unavailability due to energy saving
11	L.ChMeas.MIMO.PR.B.CL.Rank3	4G	MIMO	The number of used downlink PRBs in closed-loop rank 3 mode
12	UeCqiAvg_4G	4G	Channel Quality	The average CQI reported by the UE for the cells in a BS
13	L.ChMeas.DFEE.CarrierOFF.Dur	4G	Energy Saving	The duration of intelligent power-off of carriers
14	Distance_4G_2	4G	Network Topology	The average distance to the nearest 2 neighbour BSs operating in 4G
15	Avg_UL_RSSI	4G	Channel Quality	The average RSSI of all antennas in a 4G cell sector equipment
16	L.UL.Interference.Avg	4G	Channel Quality	The average interference and noise received in a cell
17	L.RRCRedirection.E2G	4G	Channel Quality	The number of redirections from E-UTRAN to GERAN
18	L.PDCP.Tx.Disc.Trf.SDU	4G	Channel Quality	The number of discarded downlink PDCP SDUs in carried on DRBs
19	L.CellSectorEQUIP.UL.RSSI.Avg.Ant3	4G	Channel Quality	The average RSSI of antenna 3 in a set of cell sector equipment
20	L.HHO.IntNB.IntraFreq.SuccOut.VoIP	4G	Channel Quality	The number of successful inter-eNodeB intra-freq out HO for voice UEs
21	L.RRC.ConnSetup.TimeAvg	4G	Channel Quality	The average RRC connection setup duration
22	N.ChMeas.MIMO.DL.Pair.Layer	5G	MIMO	The total number of layers that are paired for downlink MIMO
23	L.CellSectorEQUIP.UL.RSSI.Avg.Ant1	4G	Channel Quality	The average RSSI of antenna 1 in a set of cell sector equipment
24	Duplex_MODE	5G	Network Topology	The duplex mode used in the management of downlink and uplink channels
25	McsDLAvg_5G	5G	Channel Quality	The average MCS index value used in 5G for scheduling in the downlink direction
26	N.ThpVol.DL.CellOverlap	5G	Channel Quality	The volume of downlink traffic within the overlapping area in a cell
27	McsULAvg_5G	5G	Channel Quality	The average MCS index value used in 5G for scheduling in the uplink direction
28	Distance_5G_2	5G	Network Topology	The average distance to the nearest 2 neighbour BSs operating in 5G
29	N.ChMeas.MIMO.DL.Transmission.Layer	5G	MIMO	The number of transmission layers for downlink MIMO
30	Cells_Sharing_Ratio_5G	5G	Network Topology	The percentage of 5G cells operated by the MNO within a BS location
31	N.CCE.DL.AggLv12Num	5G	Channel Quality	The number of times 2 control blocks are aggregated in the downlink direction
32	N.CCE.UL.AggLv18Num	5G	Channel Quality	The number of times 8 control blocks are aggregated in the uplink direction
33	N.Cells_5G	5G	Network Topology	The total number of cells configured in a 5G BS
34	N.CCE.UL.AggLv16Num	5G	Channel Quality	The number of times 16 control blocks are aggregated in the uplink direction
35	UeCqiAvg_5G	5G	Channel Quality	The average CQI reported by the UE averaged per BS
36	N.CCE.DL.AggLv16Num	5G	Channel Quality	The number of times 16 control blocks are aggregated in the downlink direction
37	N.ChMeas.MIMO.DL.Trans.Layer.Max	5G	MIMO	The maximum number of transmission layers for downlink MIMO

- 1) **Identifying Key Variables** - Given the large number of PM variables collected at the BS level, implementing the ESF for optimising energy sustainability requires identifying the most influential variables associated with the ESC. This goal was accomplished based on the ML model and feature selection methodology, as presented in Section VIII.
- 2) **Grouping Variables by Their Nature** - To manage the diverse types of variables effectively and create a more interpretable context, grouping them according to their functions within the network architecture is useful. In the context of the ESF, a qualitative functional segmentation was introduced in Section VII. This segmentation proposes four characteristic groups (MIMO, Channel Quality, Energy Saving, and Network Topology) based on domain expertise and the classification of associated factors.
- 3) **Establishing Root-Cause Factors** - For the ESC optimisation, it is crucial that MNO's engineers understand how each variable impacts the desired outcome, particularly considering the interplay between these variables. This topic will be explored in the next subsection.

B. ROOT-CAUSE ANALYSIS

The qualitative assessment of root-cause links between the ESC and the considered variables in each scenario relied on two complementary analyses. The first analysis explores the impact of the top-ranked indicators on each class, based on the ranking presented in Table 6, considering each RAT and characteristic groups (MIMO, Channel Quality, Energy Saving, and Network Topology). The second analysis examines how changes in the selected variables affect transitions between classes over time.

1) 4G VARIABLES INFLUENCE IN THE ESC

The 4G top-ranked variables for each characteristic group are summarised in Table 7, which includes the original variable names and their aliases used for chart representation. This analysis involves a relative frequency comparison of the observed classes distribution for each variable, focusing

on aggregated key points in their distributions (e.g., intervals between CDF percentiles). This aggregated approach enhances interpretability and facilitates comparisons between variables.

TABLE 7. Variables considered in the 4G analysis.

#	Group	Original Variable	Chart Label
1)	Channel Quality	L.Traffic.DL.SCH.256QAM.DRB.TB	Traffic 256QAM 4G
2)	MIMO	L.ChMeas.MIMO.PR.B.CL.Rank2	MIMO PRB Rk2 4G
3)	Network Topology	Cells_Sharing_Ratio_4G	RAN Sharing 4G
4)	Energy Saving	L.Cell.Unavail.Dur.EnergySaving	Energy Saving 4G

The results are presented in Fig 14, with each chart corresponding to the variables numbered in Table 7.

An examination of Fig. 14-1) shows that traffic carried using higher-order downlink MCS (256 QAM), indicating good wireless channel quality, is predominantly associated with class "A" and, to a lesser extent, class "C", as can be observed by the higher percentile intervals. In contrast, classes "B" and particularly "D" are linked to lower utilisation of this higher-order downlink MCS, corresponding to the lower percentile intervals. The comparison between the intervals between the 25%-50% and 50%-75% percentiles reveals that the likelihood of a BS being classified as "A" increases from approximately 15% to around 40%, highlighting the impact of wireless channel quality on the ESC. This indicates that classes "B" and particularly the class "D", likely present low QoS in the wireless channel, hampering the utilisation of the higher order MCS. Further examination of Fig. 14-1) shows that this variable longitudinally divides the ESC scatter plot, providing insights into the SE of the analysed BSs. This behaviour is expected because higher-order MCS allow encoding more bits per symbol than lower-order schemes, signifying that more data can be transmitted per time unit using the available bandwidth, thereby increasing the SE. Moreover, the choice

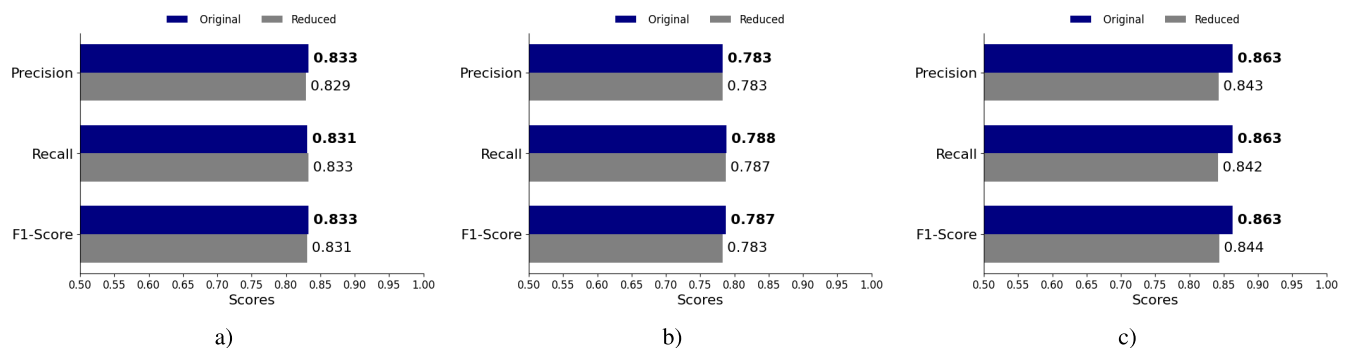


FIGURE 13. Model performance of original vs reduced feature sets: a) 4G scenario; b) 5G scenario; c) 4G/5G combined scenario.

of the downlink traffic variable by the feature selection process is anticipated, considering that, as discussed in Section V-B, downlink traffic is predominant compared to uplink traffic.

In Fig. 14-2), a similar relationship is observed between the ESC and the utilisation of MIMO in rank 2, which involves the use of two MIMO spatial streams in 4G. This relationship suggests that in 4G, utilising MIMO enhances both SE and EE. This outcome is expected, as using more spatial streams allows more data to be transmitted per unit of time without increasing the bandwidth, improving SE. Additionally, since 4G antennas are generally passive, increasing the number of spatial streams in MIMO does not lead to higher energy consumption, leading to an increase of EE.

In Fig. 14-3), it can be observed that RAN sharing with other MNOs increases the probability of observing the ESC in classes “A” and “B”, while BSs without RAN sharing tend to exhibit a higher predominance of classes “C” and “D”. This outcome is expected because shared equipment configurations typically lead to synergies in combined energy consumption. Notably, in non-sharing configurations, the probability of falling into class “D” is more than twice that in sharing scenarios, exceeding 50% of the observations. Conversely, the likelihood of observing a BS in class “A” is approximately twice as high in sharing configurations compared to non-sharing ones, reaching nearly 40% of the observations.

Fig. 14-4 illustrates the effect of energy-saving functionalities, highlighting the relationship between the classes and the activation of “sleep mode” at BS level, thus temporarily switching off the RRUs of specific cells. As expected, the results show that enabling this functionality enhances the ESC, driving BSs into class “A”. Conversely, BSs where this functionality is not used predominantly fall into classes “C” and “D”, with a likelihood approximately twice as high compared to scenarios where the functionality is activated.

2) 5G VARIABLES INFLUENCE IN THE ESC

Similarly to the 4G analysis, the 5G top-ranked variables for each characteristic group are summarised in Table 8, which includes the variable names and the respective aliases. While the top-ranked indicators in 4G encompass examples from all the considered characteristic groups, the 5G scenario includes variables only from the first three groups. This indicates that energy-saving features were given lower priority in the ESC prediction for 5G, as they are absent from the reduced selection of 37 variables shown in Table 6. Fig. 15 depicts the analysis for the three top-ranked features considered in the 5G scenario, following the order presented in Table 8.

Fig. 15-1) shows the association between the classes and the MCS used in 5G. This analysis utilises average MCS index values, which are converted to their corresponding MCS levels using the table provided in [56]. The figure

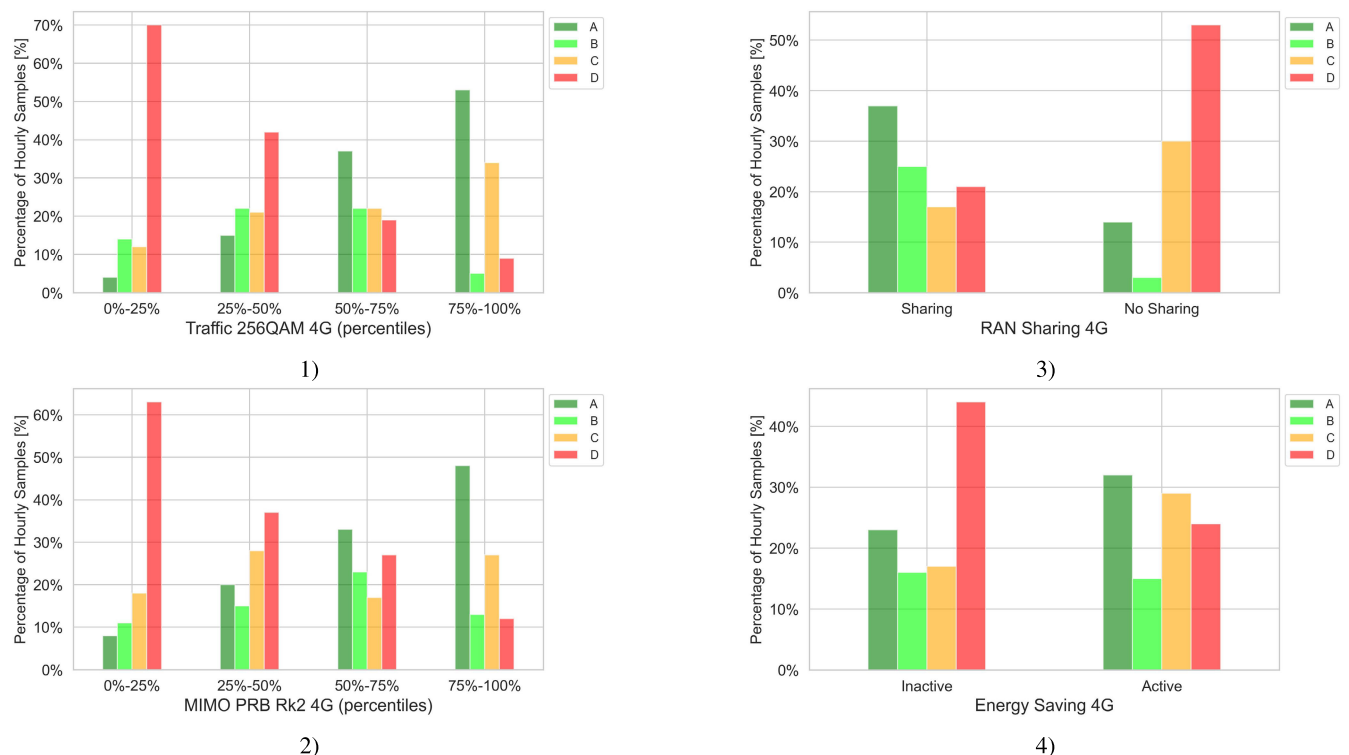


FIGURE 14. Variable influence analysis on ESC for the 4G scenario: 1) Traffic using 256 Quadrature Amplitude Modulation (QAM) (downlink); 2) Number of PRBs using two MIMO spatial streams; 3) RAN sharing status; 4) Energy-saving utilisation.

TABLE 8. Variables considered in the 5G analysis.

#	Group	Original Variable	Chart Label
1)	Channel Quality	McsDLAvg_5G	McsDLAvg 5G
2)	MIMO	N.ChMeas.MIMO.DL.Pair.Layer	MIMO Tx Layers 5G
3)	Network Topology	Duplex_MODE	Duplex Mode 5G

indicates that the use of a 64 QAM MCS increases the probability of a BS falling into class “A” by approximately four-fold compared to using a 16 QAM MCS. This finding is consistent with the earlier analysis for 4G, highlighting the positive effect of higher-order MCSs on SE. Notably, the data considered does not include any samples using Quadrature Phase Shift Keying (QPSK), which may be attributed to the inter-RAT parameterisation between 4G and 5G, which restricts 5G use to scenarios with good wireless channel quality. Moreover, no samples were observed in 5G using 256 QAM, which may indicate UE support limitations or other specific network configurations.

Fig. 15-2) reveals that the use of MIMO in 5G technology affects the classes differently from the 4G case, tending to push the BSs into classes “C” and “D”, as observed in the higher percentile intervals. This is likely a direct consequence of the Active Antenna Systems (AASs) used in 5G, particularly in the TDD mode associated with the 3.6 GHz band. Notably, while 4G generally employs passive antennas, 5G AASs integrate radio transceiver components with antenna elements. Each antenna element has active components, such as amplifiers and phase shifters, requiring individual energy feeding to control advanced functionalities like massive MIMO and beam-forming. However, the results indicate that at the current 5G utilisation level, the increase in energy consumption is not matched by a corresponding increase in the BS traffic, thus penalising the EE as shown in Equation (4).

Lastly, Fig. 15-3) depicts the association between the classes and the duplex mode configuration used the 5G

BSs. This analysis confirms that 5G TDD mode BSs (using AASs) denote a higher probability of falling into the “C” and “D” classes due to the imbalance between the extra energy consumption and the carried traffic, as discussed in Section III.

3) TIME DEPENDENCY ANALYSIS

This analysis focuses on the dynamic behaviour of the ESC over time and its interaction with key indicators in both 4G and 5G. This behaviour is illustrated in Fig. 16, where the upper figures correspond to 4G and the lower ones to 5G. As an exploratory analysis, a MIMO-related feature was selected for each: the number of PRBs using two spatial streams in 4G and the number of MIMO layers used in 5G. Two visualizations are presented for each RAT: one showing the evolution of hourly indicators and the BSs’ class over 48 hours, and another extending this analysis to the entire observation period by averaging the considered indicators over each hour. For illustrative purposes, a random BS was selected in the former analysis, while in the latter all BSs in the dataset were considered. Moreover, in this analysis, numerical values were assigned to each class (“A”=4; “B”=3; “C”=2; “D”=1). In Figs. 16-1) and 16-3), it is evident that MIMO utilisation is lower during the night period and peaks at the end of the working day hours. This pattern likely corresponds to the reduced user activity at night, leading to a lower utilisation of MIMO. Furthermore, it is observed in these 4G and 5G examples that the ESC tends to exhibit the highest class (“A”) when MIMO utilisation is higher. In Figs. 16-2) and 16-4), the analysis reveals that the statistical correlation between the ESC and MIMO utilisation is 0.55 in 4G, but only reaches 0.22 in 5G. Unlike the 4G case, where there is a relevant correlation between the two indicators, in 5G there are several periods throughout the average 24-hour period where the variation in MIMO utilisation shows an inverse relation with the BSs’ classes. This confirms the behaviour outlined in the preceding analyses regarding the impact of AASs in 5G, justifying the application of energy-saving functionalities to Optimise AAS utilisation, particularly during off-peak periods.

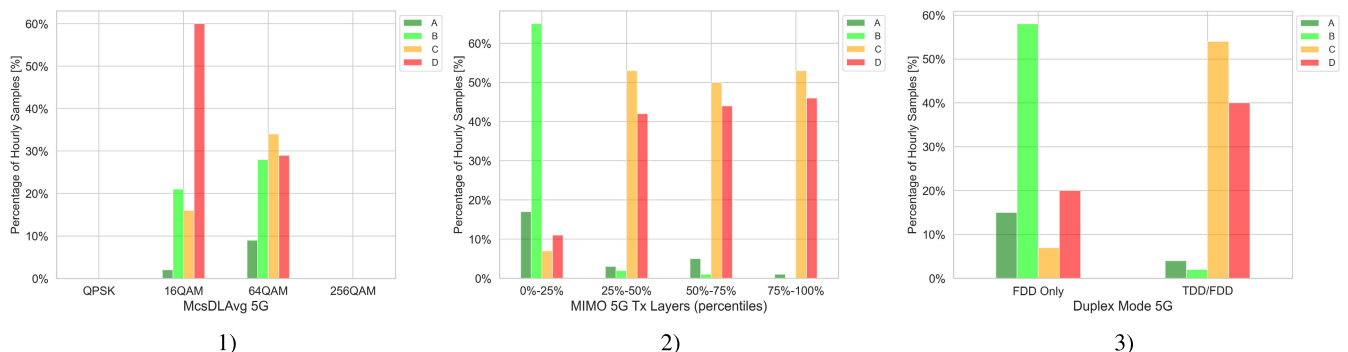


FIGURE 15. Variable influence analysis on BSs’ classes for the 5G scenario: 1) MCS utilisation in downlink; 2) Number of MIMO layers paired for downlink; 3) Duplex mode used in the 5G BSs.

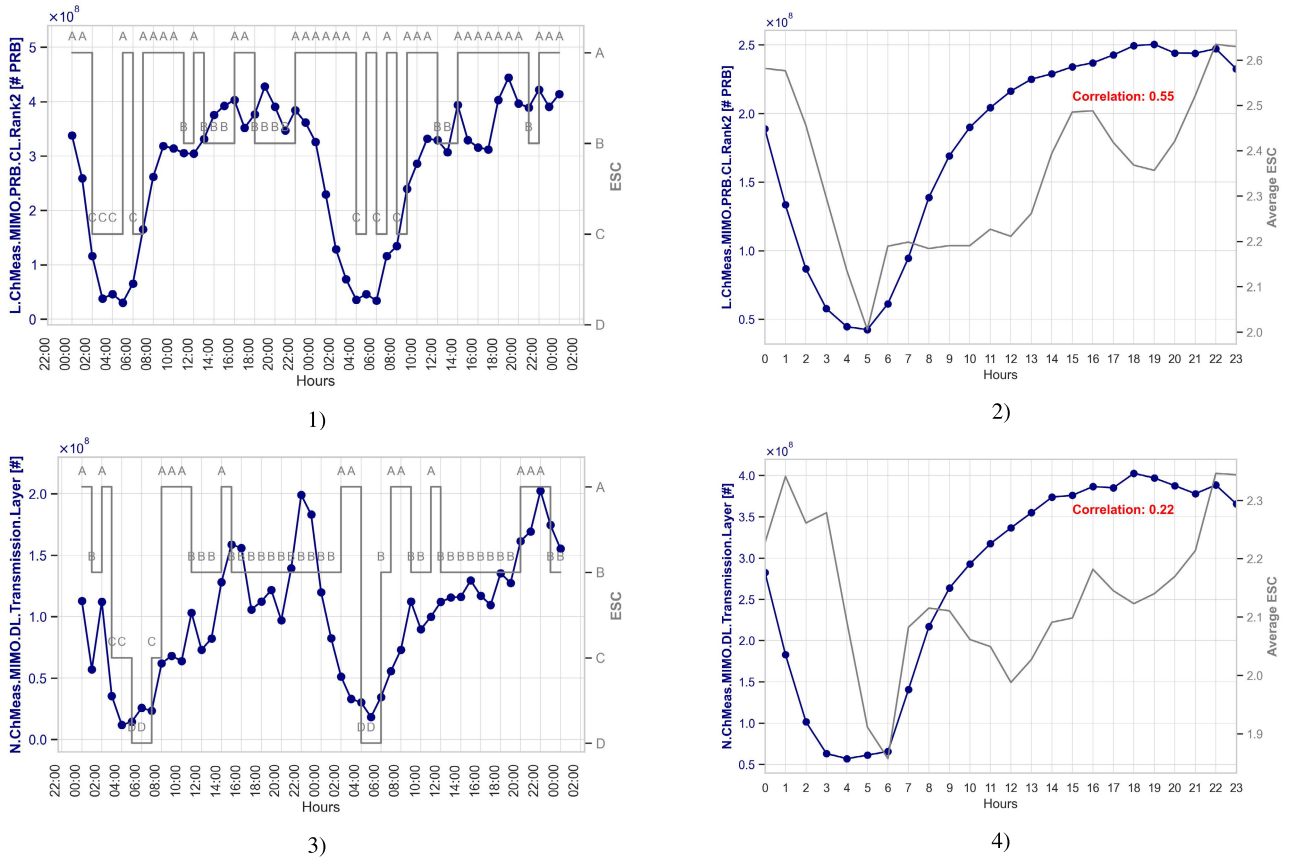


FIGURE 16. Evolution of MIMO utilisation in 4G and 5G and its impact on the BSs' class: 1) 48-hour evolution for a selected BS in 4G; 2) Average hourly evolution for all the BSs in 4G; 3) 48-hour evolution for a selected BS in 5G; 4) Average hourly evolution for all the BSs in 5G.

C. DISCUSSION

The preceding analyses delve into various perspectives on key variables identified by the ML process and their impact on the ESC. These observations confirm the existence of statistical associations between these variables and the class. However, it is important to note that statistical association does not imply causation; a third variable may be the common cause of both observed variables, as discussed in [57]. Considering the identified associations, expert knowledge is essential to select the optimal actions to enhance the BS's ESC. This expertise is crucial in interpreting the results of ML-based analysis, where statistical associations are uncovered but cannot be directly extrapolated to infer causal relationships between the model output and observed features. However, fully understanding causal relationships through expert knowledge alone can be challenging. Even if a causal relationship exists between two statistically associated variables, it is essential to determine the direction of this relationship – whether one variable causes the other or vice versa.

A crucial next step in optimising the ESC is to validate the causal relationships between the ESC and other observed PM/EM/Configuration Management (CM) indicators. This can be achieved by using statistical tools to construct Directed Acyclic Graphs (DAGs) that capture these causal relationships, guided by domain expertise and supplemented by

parametric tests as outlined in [58], or through observational causal inference techniques, as explored in [59] and [60].

X. CONCLUSION AND FUTURE WORK

Given the importance of MNOs achieving carbon neutrality by 2030, this work introduces a composite metric featuring the ESC to assist MNOs in transitioning from a load-centric optimisation paradigm, focused on maximising SE only, to a service-centric paradigm that balances SE and EE more effectively. The ESC involves the definition of a normalised metric - comprising four classes: "A"; "B"; "C" and "D" - that facilitates network auditing, benchmarking activities, and monitoring the progress of optimisation actions across the SE and EE domains. Moreover, a framework for optimising the ESC is proposed, encompassing a mix of ML and XAI techniques to pinpoint the most influential factors conditioning the ESC, thus paving the way for devising optimisation actions. These techniques are agnostic to the used ML model or RAT, allowing to aggregate results by BSs, location, equipment, or others.

Leveraging data from a real 4G/5G network, a framework combining a ML model (LGBM) with XAI techniques (Shapley values) was developed to enhance the deterministic computation of the ESC. This framework not only predicts the ESC but also pinpoints the most influential PM/EM/CM

indicators impacting the classification. The ML results, using a reduced feature set, showed an F1-Score ranging from 0.78 for the 5G scenario to 0.83 for the 4G scenario. An additional analysis, based on domain expertise, provided substantial evidence of potential causal relationships between the BSs' classes and specific variables, notably involving channel quality, MIMO utilisation, network topology and energy-saving indicators. For example, the results show that the quality of the wireless channel, denoted by the utilisation of higher order MCS, reduces by 50% the probability of a BS falling into class "D". Similarly, RAN sharing with other MNOs doubles the probability of BS being in classes "A" and "B" compared to non-sharing scenarios.

Emerging visions for 6G networks highlight the integration of AI/ML as native technologies to optimise resource allocation, enable self-healing capabilities, and enhance network intelligence. The ESF is well-positioned as a key enabler in supporting the standardisation of future energy-efficiency-aware architectures. In alignment with these objectives, future work will focus on applying observational causal inference techniques to quantify potential causal relationships and expanding the framework to larger datasets. These datasets should include a diverse range of BSs and environments, such as dense urban, urban, suburban, and rural scenarios, to evaluate generalisability. Additionally, integrating causal inference methods with Large Language Models (LLMs) presents an important research direction for advancing wireless network applications and further supporting the development of energy-efficient architectures for 6G.

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